File

Name: M39-baseline-K03-premachining
Description: KISSsoft example
Changed by: AAA on: 31.10.2026 at: 03:45:45

1 Overview

Calculation method		ISO 6336:2019	
Drawing or article number:			
Gear 1:		0.000.0	
Gear 2:		0.000.0	
		------- Gear 1	---------- Gear 2 -------
Power (kW)	[P]	8.671	
Speed (1/min)	[n]	1200.0	320.0
Number of load cycles (in mio.)	[NL]	72.000	19.200
Torque (Nm)	[T]	69.0	258.8
Application factor	[KA]	1.10	
Required service life (h)	[H]	1000.00	
Gear driving (+) / driven (-)		+	-
Working flank Gear 1:		Right flank	
Gear 1 direction of rotation:		Clockwise	

2 Tooth geometry

Geometry calculation according to		ISO 21771:2007 (withdrawn)	
		------- Gear 1	---------- Gear 2 -------
Center distance (mm)	[a]	72.500	
Center distance tolerance		ISO 286:2010 Measure js7	
Normal module (mm)	[mn]	2.5000	
Transverse module (mm)	[mt]	2.5000	
Normal diametral pitch (1/in)	[Pnd]	10.16000	
Transverse diametral pitch (1/in)	[Ptd]	10.16000	
Normal pressure angle (°)	[αn]	20.0000	
Helix angle at reference circle (°)	[β]	0.0000	
Number of teeth	[z]	12	45
Facewidth (mm)	[b]	50.00	48.00
Hand of gear		Spur gear	
Accuracy grade	[Q-ISO 1328:2013]	A6	A7
Inner diameter (mm)	[di]	12.00	84.06
Shaft diameter (mm)	[ds]	12.00	84.06
Inner diameter of gear rim (mm)	[dbi]	0.00	0.00
		------- Gear 1	---------- Gear 2 -------

3 Materials

Gear 1

18CrNiMo7-6, Case hardening steel, case-hardened, ISO 6336-5 Figure 9/10 (MQ), Core hardness ≥25HRC Jominy J=12mm<HRC28

Gear 2

34CrAlMo5-10, Nitriding steel, gas-nitrided, ISO 6336-5 Figure 13a/14a (MQ)

		------- Gear 1 ----------	Gear 2 -------
Surface hardness		HRC 61	HV 950
Material treatment according to ISO 6336:2006 Normal, Life factors Z_{NT} and Y_{NT} ≥ 0.85			
Infinite life strength for tooth root stress (N/mm²)	[σFlim]	425.00	420.00
Infinite life strength for Hertzian pressure (N/mm²)	[σHlim]	1500.00	1250.00
Young's modulus (N/mm²)	[E]	206000	206000
Poisson's ratio	[v]	0.300	0.300
Tensile strength (N/mm²)	[σB]	1200.00	800.00
Yield point (N/mm²)	[σS]	850.00	600.00

3.1 Gear roughness

		------- Gear 1 ----------	Gear 2 -------
Arithmetic mean roughness value R_a, flank (µm)	[RAH]	0.60	2.00
Arithmetic mean roughness value R_a, root (µm)	[RAF]	3.00	2.00
Mean peak-to-valley roughness R_z, flank (µm)	[RZH]	4.80	14.00
Mean peak-to-valley roughness R_z, root (µm)	[RZF]	20.00	14.00

3.2 Lubrication

Lubrication type	Oil bath lubrication	
Type of oil	ISO-VG 220	
Lubricant base	Mineral-oil base	
Lubricant factor	[XL]	1.000
Oil nominal kinematic viscosity at 40°C (mm²/s)	[v40]	220.00
Oil nominal kinematic viscosity at 100°C (mm²/s)	[v100]	17.50
Specific density at 15°C (kg/dm³)	[ρ]	0.895
Oil temperature (°C)	[TS]	70.000

4 Geometry

4.1 Reference profiles

4.1.1 Information on pre-machining

Tool reference profile from Gear 1

Cutter label, own input		
Min. finishing stock (mm)	[qmin]	0.110
Addendum coefficient	[haP0*]	1.386
Dedendum coefficient	[hfP0*]	1.200
Tip radius factor	[ρaP0*]	0.200
	[ρaP0max*]	0.401
Root radius factor	[ρfP0*]	0.001
Protuberance height coefficient	[hprP0*]	0.000
Protuberance angle (°)	[αprP0]	0.000
Root form height coefficient	[hFfP0*]	0.000
Ramp angle (°)	[αKP0]	0.000
Tooth thickness coefficient reference line	[sP0*]	1.571
	not topping	
Addendum coefficient reference profile Gear	[haP*]	1.000
Manufacturing center distance (mm)	[a0]	16.609
(mm)	[a0.e/i]	16.535 /16.494
Smallest radius of curvature, root rounding (mm)	[ρmin.e/i]	0.624 / 0.631

Tool reference profile from Gear 2

Cutter label, own input		
Min. finishing stock (mm)	[qmin]	0.100
Protuberance (mm)	[pr0]	0.150
Remaining undercut (mm)	[spr0 = pr0-q]	0.050

Addendum coefficient	[haP0*]	1.300
Dedendum coefficient	[hfP0*]	1.200
Tip radius factor	[ρaP0*]	0.250
	[ρaP0max*]	0.498
Root radius factor	[ρfP0*]	0.001
Protuberance height coefficient	[hprP0*]	0.603
Protuberance angle (°)	[αprP0]	12.000
Root form height coefficient	[hFfP0*]	0.000
Ramp angle (°)	[αKP0]	0.000
Tooth thickness coefficient reference line	[sP0*]	1.571
	not topping	
Addendum coefficient reference profile Gear	[haP*]	1.000
Manufacturing center distance (mm)	[a0]	56.679
(mm)	[a0.e/i]	56.583 /56.528
Smallest radius of curvature, root rounding (mm)	[ρmin.e/i]	0.715 / 0.719

4.1.2 Reference profile of the final tooth form

		------- Gear 1	---------- Gear 2 -------
Dedendum reference profile	[hfP*]	1.241	1.161
Tooth root radius reference profile	[ρfP*]	0.200	0.250
Addendum reference profile	[haP*]	1.000	1.000
Protuberance height coefficient	[hprP*]	0.000	0.269
Protuberance angle (°)	[αprP]	0.000	12.000
Tip form height coefficient	[hFaP*]	0.000	0.000
Ramp angle (°)	[αKP]	0.000	0.000
Data for final machining:			
Depth of immersion	[hgrind*]	1.071	0.800
Radius at cutter head	[rgrind*]	0.100	0.100
Grinding only flank (0), flank & root (1)		0	0
Generation grinding (0), form grinding (1)		0	0
Type of profile modification:	none (only running-in)		
Tip relief by running in (µm)	[Ca L/R]	2.0 / 2.0	3.2 / 3.2

4.2 Basic data

Overall transmission ratio	[itot]	-3.750
Gear ratio	[u]	3.750
Transverse module (mm)	[mt]	2.500
Transverse pressure angle (°)	[αt]	20.000
Working pressure angle (°)	[αwt]	22.558
(°)	[αwt.e/i]	22.587 /22.530
Working pressure angle at normal section (°)	[αwn]	22.558
Base helix angle (°)	[βb]	0.000
Helix angle at operating pitch circle (°)	[βw]	0.000
Reference center distance (mm)	[ad]	71.250
Sum of profile shift coefficients	[Σxi]	0.5313

		------- Gear 1	---------- Gear 2 -------
Profile shift coefficient			
Information on pre-machining	[xV]	0.6437	0.1716
Information on final machining	[x]	0.4986	0.0327
Generating profile shift coefficient			
Information on pre-machining	[xEV.e/i]	0.6140/ 0.5975	0.1332/ 0.1112
Information on final machining	[xE.e/i]	0.4689/ 0.4524	-0.0058/-0.0278
Virtual gear no. of teeth	[zn]	12.000	45.000
Involute length (mm)	[l_dFa-l_dFf]	5.346	4.388

4.3 Diameters and their allowances

			------- Gear 1	---------- Gear 2 -------
Reference diameter (mm)		[d]	30.000	112.500
Base diameter (mm)		[db]	28.191	105.715
Tip alteration (mm)		[k*mn]	0.000	-0.125
Tip diameter (mm)		[da]	37.493	117.413
(mm)		[da.e/i]	37.493 /37.483	117.413 /117.403
Tip diameter allowances (mm)		[Ada.e/i]	0.000 / -0.010	0.000 / -0.010
Tip form diameter (mm)		[dFa]	37.493	117.413
(mm)		[dFa.e/i]	37.493 /37.483	117.413 /117.403
Root form diameter, pre-machining (mm)		[dFa_p]	37.493	117.413
	(mm)	[dFa_p.e/i]	37.493 /37.483	117.413 /117.403
Active tip diameter (mm)		[dNa]	37.493	117.413
(mm)		[dNa.e/i]	37.493 /37.483	117.413 /117.403
V-Circle diameter (mm)		[dv]	32.493	112.663
(mm)		[dv.e/i]	32.344 /32.262	112.471 /112.361
Operating pitch diameter (mm)		[dw]	30.526	114.474
(mm)		[dw.e/i]	30.533 /30.520	114.497 /114.450
Root diameter (mm)		[df]	26.290	106.858
(mm)		[df.e/i]	26.142 /26.059	106.666 /106.556
(calculated with pre-machining tool)				
Active root diameter (mm)		[dNf]	28.553	110.141
(mm)		[dNf.e/i]	28.569 /28.541	110.167 /110.119
Root form diameter (mm)		[dFf]	28.335	109.226
(mm)		[dFf.e/i]	28.294 /28.275	109.274 /109.052
(calculated with final machining tool)				
Root form diameter, pre-machining (mm)		[dFf_p]	28.222	110.302
	(mm)	[dFf_p.e/i]	28.205 /28.199	110.180 /110.115

4.4 Tip clearances and tooth heights

		------- Gear 1	---------- Gear 2 -------
Theoretical tip clearance (mm)	[c]	0.325	0.648
Effective tip clearance (mm)	[c.e/i]	0.496 / 0.406	0.784 / 0.707
Reserve (dNf-dFf)/2 (mm)	[cF.e/i]	0.147 / 0.123	0.557 / 0.422
Undercut taken into account			
Addendum, $m_n(h_{aP}*+x+k)$ (mm)	[ha]	3.746	2.457
(mm)	[ha.e/i]	3.746 / 3.741	2.457 / 2.452
Dedendum, $m_n(h_{fP}*-x)$ (mm)	[hf]	1.855	2.821
(mm)	[hf.e/i]	1.929 / 1.970	2.917 / 2.972
Tooth height (mm)	[h]	5.601	5.278

4.5 Roll angle

		------- Gear 1	---------- Gear 2 -------
Roll angle at dFa (°)	[ξFa.e/i]	50.238 /50.207	27.690 /27.677
Roll angle to dNf (°)	[ξNf.e/i]	9.423 / 9.058	16.801 /16.708
Roll angle at dFf (°)	[ξFf.e/i]	4.918 / 4.428	14.991 /14.509

4.6 Tooth thickness and pitch

		------- Gear 1	---------- Gear 2 -------
Tooth thickness on reference circle, arc, in module	[sn*]	1.9337	1.5946
Normal tooth thickness at tip circle (mm)	[san]	0.716	2.025
(mm)	[san.e/i]	0.657 / 0.611	1.956 / 1.910
Normal tooth thickness at tip form circle (mm)	[sFan]	0.716	2.025
(mm)	[sFan.e/i]	0.657 / 0.611	1.956 / 1.910
Normal space width at root circle (mm)	[efn]	0.000	2.194
(mm)	[efn.e/i]	0.000 / 0.000	2.229 / 2.250
Pitch on reference circle (mm)	[pt]	7.854	7.854
Base pitch (mm)	[pbt]	7.380	7.380
Transverse pitch on contact-path (mm)	[pet]	7.380	7.380

4.7 Sliding

		------- Gear 1	---------- Gear 2 -------
Max. sliding velocity at tip (m/s)	[vga]	1.035	0.571
Specific sliding at the tip	[ζa]	0.667	0.667
Specific sliding at the root	[ζf]	-2.004	-1.999
Mean specific sliding	[ζm]	0.667	
Sliding factor on tip	[Kga]	0.540	0.298
Sliding factor on root	[Kgf]	-0.298	-0.540

4.8 Contact ratios

		---------- Pair ----------
Minimal length of contact line (mm)	[Lmin]	48.000
Transverse contact ratio	[εα]	1.367
	[εα.e/m/i]	1.373 / 1.366/ 1.359
Overlap ratio	[εβ]	0.000
Total contact ratio	[εγ]	1.367
	[εγ.e/m/i]	1.373 / 1.366/ 1.359
Length of path of contact (mm)	[gα]	10.092
(mm)	[gα.e/m/i]	10.131 /10.082/10.033
Approach path of contact (mm)	[gf]	3.588
(mm)	[gf.e/m/i]	3.619 / 3.582/ 3.545
Recess path of contact (mm)	[ga]	6.504
(mm)	[ga.e/m/i]	6.512 / 6.500/ 6.488

		------- Gear 1	---------- Gear 2 -------
Addendum contact ratio	[ε]	0.881	0.486
	[ε.e/i]	0.882 / 0.879	0.490 / 0.480
Length T1-A and T2-A (mm)	[T1A,T2A]	2.268	25.545
(mm)	[.e/i]	2.228 / 2.318	25.545 / 25.533
Length T1-B and T2-B (mm)	[T1B,T2B]	4.979	22.834
(mm)	[.e/i]	4.979 / 4.971	22.795 / 22.880
Length T1-C and T2-C (mm)	[T1C,T2C]	5.855	21.957
(mm)	[.e/i]	5.847 / 5.863	21.926 / 21.988

Length T1-D and T2-D (mm)	[T1D,T2D]	9.648	18.165
(mm)	[.e/i]	9.609 / 9.698	18.165 / 18.153
Length T1-E and T2-E (mm)	[T1E,T2E]	12.359	15.453
(mm)	[.e/i]	12.359 / 12.352	15.414 / 15.500
Length T1-T2 (mm)	[T1T2]	27.812	
(mm)	[.e/i]	27.773 / 27.852	
Diameter of single contact point B (mm)	[d-B]	29.898	115.158
(mm)	[d-B.e/i]	29.898 / 29.893	115.127 /115.195
Diameter of single contact point D (mm)	[d-D]	34.162	111.784
(mm)	[d-D.e/i]	34.118 / 34.219	111.784 /111.776

5 General influence factors

5.1 Forces and circumferential speed

		------- Gear 1	---------- Gear 2 -------
Nominal circum. force at pitch circle (N)	[Ft]	4600.00	
Axial force (N)	[Fa]	0.00	0.00
Radial force (N)	[Fr]	1674.26	1674.26
Normal force (N)	[Fnorm]	4895.22	4895.22
Nominal circumferential force per mm (N/mm)	[w]	95.83	
Only as information: Forces at operating pitch circle:			
Nominal circumferential force (N)	[Ftw]	4520.69	
Axial force (N)	[Faw]	0.00	0.00
Radial force (N)	[Frw]	1877.90	
Circumferential speed reference circle (m/s)	[v]	1.88	
Circumferential speed operating pitch circle (m/s)	[v(dw)]	1.92	

5.2 Contact stiffness

Running-in value (µm)	[yp]	0.8
Running-in value (µm)	[yf]	1.0
Correction factor	[CM]	0.800
Gear blank factor	[CR]	1.000
Basic rack factor	[CB]	1.000
Material coefficient	[E/Est]	1.000
Single stiffness (N/mm/µm)	[c']	13.556
Theoretical single stiffness (N/mm/µm)	[c'th]	16.952
Contact stiffness (N/mm/µm)	[cγα]	17.291
Contact stiffness (N/mm/µm)	[cγβ]	14.697
Reduced mass (kg/mm)	[mRed]	0.00347
Resonance speed (min-1)	[nE1]	56211
Resonance ratio (-)	[N]	0.021
Subcritical range		

5.3 Calculation of K factors

Running-in value (µm)	[yα]	1.0	
Bearing distance l of pinion shaft (mm)	[l]	100.000	
Distance s of pinion shaft (mm)	[s]	10.000	
Outside diameter of pinion shaft (mm)	[dsh]	27.270	
Load in accordance with Figure 13, ISO 6336-1:2006 0:a), 1:b), 2:c), 3:d), 4:e)	[-]		4
Coefficient K' according to Figure 13, ISO 6336-1:2006 Without stiffening	[K']		-1.00
Effective flank line deviation (µm)	[Fβy]	5.95	

from deformation of shaft (µm)	[fsh·B1]	7.81
fsh (µm)	[fsh]	7.81
B1	[B1]	1.00
fHβ5 (µm)	[fHβ5]	7.00
Tooth without flank line modification		
Position of contact pattern:	favorable	
from production tolerances (µm)	[fmα*B2]	16.92
B_2=1.00		
Tooth trace deviation, theoretical (µm)	[Fβx]	7.00
Running-in value (µm)	[yβ]	1.05

5.4 K factors

Dynamic factor	[Kv]	1.025
Face load factors		
- Flank	[KHβ]	1.405
- Tooth root	[KFβ]	1.353
- Scuffing	[KBβ]	1.405
Transverse load factors		
- Flank	[KHα]	1.000
- Tooth root	[KFα]	1.000
- Scuffing	[KBα]	1.000
Application factor	[KA]	1.100
Mesh load factor	[Kγ]	1.000

6 Calculation of tooth root strength (fracture)

Calculation of Tooth form coefficients according method: B

		------- Gear 1 ----------	---------- Gear 2 -------
Calculated with generating profile shift coefficient	[xE.i]	0.5975	-0.0278
Tooth form factor	[YF]	1.568	1.833
Stress correction factor	[YS]	2.264	1.974
Load application angle (°)	[αFen]	29.291	21.923
Load distribution influence factor	[fε]	1.000	
Load application diameter (mm)	[d_{en}]	34.162	115.158
Bending moment arm (mm)	[hFe]	3.176	3.388
(-)	[hFe*]	1.270	1.355
Tooth root thickness at critical cross section (mm)	[sFn]	5.310	5.232
(-)	[sFn*]	2.124	2.093
Root fillet radius at critical cross section (mm)	[ρF]	0.791	1.036
(-)	[ρF*]	0.316	0.414
Tangent contact point (mm)	[x,y]	2.655 /12.986	
Tangent contact point (mm)	[x,y]		2.616 /53.590
Diameter at critical cross section (mm)	[d_{sFn}]	26.509	107.308
Tangent at critical cross section (°)	[α_{sFn}]	30.000	30.000
Notch parameter	[q_s]	3.357	2.525
Helix angle factor	[Yβ]	1.000	
Deep tooth factor	[YDT]	1.000	
Gear rim factor	[YB]	1.00	1.00
Effective facewidth (mm)	[beff]	50.00	48.00
Nominal stress at tooth root (N/mm²)	[σF0]	130.65	138.70
Tooth root stress (N/mm²)	[σF]	199.32	211.59
Permissible bending stress at root of Test-gear			
Notch sensitivity factor	[YδrelT]	1.008	1.001

Surface factor	[YRrelT]	0.957	0.996
Size factor, tooth root	[YX]	1.000	1.000
Life factor	[YNT]	0.938	0.963
$Y_{\delta relT} \cdot Y_{RrelT} \cdot Y_X \cdot Y_{NT}$		0.905	0.960
Alternating bending factor, mean stress influence coefficient	[YM]	1.000	1.000
Stress correction factor	[YST]	2.00	
$Y_{ST} \cdot \sigma_{Flim}$ (N/mm²)	[σFE]	850.00	840.00
Permissible tooth root stress σFG/SFmin (N/mm²)	[σFP]	549.18	576.16
Limit strength tooth root (N/mm²)	[σFG]	768.85	806.62

6.1 Safety factors

		------- Gear 1	---------- Gear 2 -------
Required safety	[SFmin]	1.40	1.40
Safety for tooth root stress	[SF=σFG/σF]	3.86	3.81
Transmittable power (kW)	[kWRating]	23.89	23.61

7 Calculation of flank strength (pitting)

		------- Gear 1	---------- Gear 2 -------
Zone factor	[ZH]	2.335	
Elasticity coefficient (√N/mm)	[ZE]	189.812	
Contact ratio factor	[Zε]	0.937	
Helix angle factor	[Zβ]	1.000	
Effective facewidth (mm)	[beff]	48.00	
Nominal contact stress (N/mm²)	[σH0]	835.18	
Contact stress at operating pitch circle (N/mm²)	[σHw]	1051.16	
Single tooth contact factor	[ZB,ZD]	1.06	1.00
Contact stress (N/mm²)	[σHB, σHD]	1117.85	1051.16
Lubricant coefficient for N_L	[ZL]	1.020	1.020
Speed factor at N_L	[ZV]	0.963	0.963
Roughness factor for N_L	[ZR]	0.894	0.894
Material hardening factor for N_L	[ZW]	1.000	1.000
Life factor	[ZNT]	0.989	0.958
Size factor (flank)	[ZX]	1.000	1.000
$Z_L \cdot Z_V \cdot Z_R \cdot Z_{NT} \cdot Z_X$		0.869	0.841
Limited pitting is permissible:	No		
Permissible contact stress, σHG/SHmin (N/mm²)	[σHP]	1302.97	1051.64
Pitting stress limit (N/mm²)	[σHG]	1302.97	1051.64

7.1 Safety factors

		------- Gear 1	---------- Gear 2 -------
Required safety	[SHmin]	1.00	1.00
Safety factor for contact stress on operating pitch circle	[SHw]	1.24	1.00
Safety against pressure, σHG/σHBD Single contact	[SHBD]	1.17	1.00
Safety regarding transmittable torque	[SHBD²]	1.36	1.00
Transmittable power (kW)	[kWRating]	11.78	8.68

8 Micropitting

Calculation method according to ISO/TS 6336-22:2018

Calculation has not been carried out, lubricant: Load stage micropitting test not known

9 Scuffing load capacity

Calculation method according to ISO/TS 6336-20/21:2022

		------- Gear 1	---------- Gear 2 -------
Thermal contact coefficient (N/mm/s^.5/K)	[BM]	13.780	13.780
Relevant tip relief (µm)	[Ca]	2.00	3.20
Optimal tip relief (µm)	[Ceff]	7.78	
C_a taken as optimal in the calculation. 0=no, 1=yes		0	0

		---------- Pair ----------	
Helical load factor for scuffing	[KBγ]	1.000	
Lubrication coefficient for lubrication type	[XS]	1.000	
Scuffing test, load stage	[FZGtest]	FZG - Test A / 8.3 / 90 (ISO 14635 - 1),	12
Multiple meshing factor	[Xmp]	1.000	
Relative structural factor, scuffing	[XWrelT]	1.000	
Effective facewidth (mm)	[beff]	48.000	
Applicable circumferential force/facewidth (N/mm)	[wBt]	151.808	
$K_{B\gamma}$ = 1.000, $w_{Bt} \cdot K_{B\gamma}$ = 151.808			
Angle factor	[Xαβ]	1.015	
ε_1 =0.881, ε_2 =0.486			

9.1 Flash temperature-criteria

		---------- Pair ----------	
Lubricant factor	[XL]	0.830	
Tooth mass temperature (°C)	[θMi]	76.43	
$\theta_{Mi} = \theta_{oil} + X_S \cdot 0.47 \cdot X_{mp} \cdot \theta_{flm}$			
Average flash temperature (Δ°C)	[θflm]	13.68	
Scuffing temperature (°C)	[θS]	348.81	
Γ coordinates (point of highest temperature)	[Γ]	0.641	
[Γ.A]= -0.613, [Γ.E]= 1.111			
Maximum flash temperature (Δ°C)	[θflamax]	26.05	
Maximum contact temperature (°C)	[θBmax]	102.48	
Flash factor (°K·N^-.75·s^.5·m^-.5·mm)	[XM]	50.058	
Approach factor	[XJ]	1.000	
Load sharing factor	[XΓ]	1.000	
Dynamic viscosity (mPa*s)	[ηM]	41.87	(70.0 °C)
Coefficient of friction	[μ_m]	0.087	
Required safety	[SBmin]	2.000	
Margin of safety for scuffing, flash temperature	[SB]	8.584	

9.2 Integral temperature-criteria

		---------- Pair ----------	
Lubricant factor	[XL]	1.000	
Tooth mass temperature (°C)	[θMC]	77.99	
$\theta_{MC} = \theta_{oil} + X_S \cdot 0.70 \cdot \theta_{flaint}$			
Integral scuffing temperature (°C)	[θSint]	360.78	
Flash factor (°K·N^-.75·s^.5·m^-.5·mm)	[XM]	50.058	
Running-in factor, well run in	[XE]	1.000	
Contact ratio factor	[Xε]	0.279	
Dynamic viscosity (mPa*s)	[ηOil]	41.87	(70.0 °C)
Mean coefficient of friction	[μ_m]	0.111	
Geometry factor	[XBE]	0.444	
Meshing factor	[XQ]	1.000	
Tip relief factor	[XCa]	1.247	
Mean flash temperature (Δ°C)	[θflaint]	11.41	
Integral tooth flank temperature (°C)	[θint]	95.11	
Required safety	[SSmin]	1.800	
Safety factor for scuffing (intg.-temp.)	[SSint]	3.793	
Safety factor for transmitted moment (int.-T.)	[SSL]	11.580	

10 Measurements for tooth thickness

10.1 Tooth thickness tolerances and inputs

		------- Gear 1 ----------	Gear 2 -------
10.1.1 Information on pre-machining			
Min. finishing stock (mm)	[qmin]	0.110	0.100
Tooth thickness tolerance pre-machining (mm)	[TsnV]	0.030	0.040
Additional measure for pre-machining (mm)	[ΔAs_p.e/i]	0.294 / 0.234	0.293 / 0.213
Tooth thickness allowance (normal section) (mm)	[As_p.e/i]	0.210 / 0.180	0.183 / 0.143
10.1.2 Information on final machining			
Tooth thickness tolerance		DIN 3967 cd25	DIN 3967 cd25
Tooth thickness allowance (normal section) (mm)	[Asn.e/i]	-0.054 / -0.084	-0.070 / -0.110

10.2 Base tangent lengths

		------- Gear 1 ----------	Gear 2 -------
10.2.1 Information on pre-machining			
Number of teeth spanned	[k]	3.000	6.000
Base tangent length with allowance (mm)	[Wk.e/i]	19.921 /19.893	42.395 /42.358
(mm)	[ΔWk.e/i]	0.197 / 0.169	0.172 / 0.134
10.2.2 Information on final machining			
Number of teeth spanned	[k]	3.000	6.000
Base tangent length (no backlash) (mm)	[Wk]	19.724	42.223
Base tangent length with allowance (mm)	[Wk.e/i]	19.673 /19.645	42.158 /42.120
(mm)	[ΔWk.e/i]	-0.051 / -0.079	-0.066 / -0.103
Diameter of measuring circle (mm)	[dMWk.m]	34.368	113.804

10.3 Measurement over balls and pins

		------- Gear 1 ----------	Gear 2 -------
10.3.1 Information on pre-machining			
Effective diameter of ball/pin (mm)	[DMeff]	6.000	4.250
Diametral measurement over two balls (mm)	[MdK.e/i]	42.497 /42.452	118.836 /118.739
10.3.2 Information on final machining			
Theoretical diameter of ball/pin (mm)	[DM]	5.672	4.248
Effective diameter of ball/pin (mm)	[DMeff]	6.000	4.250
Radial single-ball measurement, no backlash (mm)	[MrK]	21.092	59.229
Radial single-ball measurement (mm)	[MrK.e/i]	21.051 /21.029	59.141 /59.091
Diameter of measuring circle (mm)	[dMMr.m]	32.674	112.462
Diametral measurement over two balls, no backlash (mm)	[MdK]	42.184	118.388
Diametral measurement over two balls (mm)	[MdK.e/i]	42.103 /42.058	118.213 /118.113
Diametral measurement over pins, no backlash (mm)	[MdR]	42.184	118.388
Measurement over pins according to DIN 3960 (mm)	[MdR.e/i]	42.103 /42.058	118.213 /118.113
Measurement over 3 pins, no backlash (mm)	[Md3R]	0.000	118.318
Measurement over 3 pins (mm)	[Md3R.e/i]	0.000 / 0.000	118.144 /118.043

10.4 Tooth thickness

		------- Gear 1	---------- Gear 2 -------
10.4.1 Information on pre-machining			
Medium tip diameter (mm)	[da.m]	37.488	117.408
Reference chordal height from da.m (mm)	[hac]	3.955	2.492
Tooth thickness at height hac, chord (mm)	[sc.e/i]	5.020 / 4.992	4.168 / 4.129
Tooth thickness on reference circle, arc (mm)	[sn.e/i]	5.044 / 5.014	4.169 / 4.129
10.4.2 Information on final machining			
Medium tip diameter (mm)	[da.m]	37.488	117.408
Reference chordal height from da.m (mm)	[hac]	3.939	2.490
Tooth thickness at height hac, chord (mm)	[sc]	4.813	3.986
(mm)	[sc.e/i]	4.762 / 4.733	3.916 / 3.877
Tooth thickness on reference circle, arc (mm)	[sn]	4.834	3.986
(mm)	[sn.e/i]	4.780 / 4.750	3.916 / 3.876

10.5 Backlash

		---------- Pair ----------
Circumferential backlash from Aa (mm)	[jtw_Aa.e/i]	0.012 / -0.012
Radial backlash (mm)	[jrw.e/i]	0.255 / 0.138
Circumferential backlash (transverse section) (mm)	[jtw.e/i]	0.210 / 0.114
Normal backlash (mm)	[jnw.e/i]	0.193 / 0.106
Center distance allowances (mm)	[Aa.e/i]	0.015 / -0.015
Backlash free center distance (mm)	[aControl.e/i]	72.347 / 72.260
Backlash free center distance, allowances (mm)	[jta]	-0.153 / -0.240

		------- Gear 1	---------- Gear 2 -------
dNf.i with aControl (mm)	[dNf0.i]	28.380	109.793
Reserve (dNf0.i-dFf.e)/2 (mm)	[cF0.i]	0.043	0.260
Tip clearance (mm)	[c0.i(aControl)]	0.181	0.483

Torsional angle Gear 2 with fixed Gear 1:			
Total torsional angle (°)	[j.tSys]	0.2101 / 0.1138	(0°12'36"/ 0°6'50")
Torsional angle Gear 1 with fixed Gear 2:			
Total torsional angle (°)	[j.tSys]	0.7878 / 0.4269	(0°47'16"/ 0°25'37")

11 Gear tolerances

		------- Gear 1	---------- Gear 2 -------
According to ISO 1328-1:2013, ISO 1328-2:1997			
Accuracy grade	[Q]	A6	A7
Single pitch tolerance (µm)	[fpT]	±8.50	±12.00
Base pitch tolerance (µm)	[fpbT]	±7.99	±11.28
Sector pitch tolerance (µm)	[Fpz/8T]	±19.00	±27.00
Cumulative pitch tolerance, total (µm)	[FpT]	24.00	40.00
Adjacent pitch difference tolerance (µm)	[fuT]	12.00	17.00
Profile form tolerance (µm)	[ffαT]	9.00	13.00
Profile slope tolerance (µm)	[fHαT]	±7.00	±10.00
Profile tolerance, total (µm)	[FαT]	11.00	16.00
Helix form tolerance (µm)	[ffβT]	11.00	16.00
Helix slope tolerance (µm)	[fHβT]	±9.50	±14.00
Helix tolerance, total (µm)	[FβT]	14.00	21.00
Runout tolerance (µm)	[FrT]	21.00	36.00

Single flank composite tolerance, total (µm)	[FisT]	32.00	51.00
Single flank composite tolerance, tooth-to-tooth (µm)	[fisT]	8.50	12.00
Radial composite tolerance, total (µm)	[Fi"]	26.00	43.00
Radial composite tolerance, tooth-to-tooth (µm)	[fi"]	9.50	13.00

F_{idT} (F_i"), f_{idT} (f_i") according to ISO 1328-2:1997 calculated with the geometric mean values for m_n and d.

According to ISO 1328-2:2020

Accuracy grade	[R]	R39	R41
Radial composite tolerance, total (µm)	[FidT]	28.00	43.00
Radial composite tolerance, tooth-to-tooth (µm)	[fidT]	16.00	18.00

Axis alignment tolerances recommendation according to ISO/TR 10064-3:1996, AGMA 915-3-A99

Total helix deviation for calculation (µm)	[Fβ]	21.00
Distance between bearings for calculation (mm)	[LG]	100.00
Maximum value for axis deviation error (µm)	[fΣβ]	± 21.88
Maximum value for axis inclination error (µm)	[fΣδ]	± 43.75

12 Modifications and determination of the tooth form

12.1 Data for the tooth form calculation

Data not available.
Please run the calculation in the "Tooth form" tab and open the main report again.

13 Supplementary data

Maximal possible centre distance (ε_α = 1.0)	[aMAX]	73.583

13.1 Masses, stiffnesses and moments of inertia

		------- Gear 1	---------- Gear 2 -------
Mass (kg)	[m]	0.268	1.626
Total mass (kg)	[Σm]	1.895	
Moment of inertia for system, relative to the input:			
calculation without consideration of the exact tooth shape			
Gears individually ((da+df)/2...di) (kg*m²)	[J]	3.896e-05	0.003992
System (da+df)/2...di (kg*m²)	[J]	0.0003229	

13.2 Wear, power loss, sound pressure level

Torsional stiffness at driving gear with fixed driven gear:

Torsional stiffness (MNm/rad)	[cr]	0.138
Torsion when subjected to nominal torque (°)	[δcr]	0.029
Sound pressure level based on Masuda, without PPTE/δs	[dB(A)]	61.131

13.3 Power loss

Average coefficient of friction according to Niemann	[μ_m]	0.114
Wear sliding coefficient by Niemann	[ζw]	0.912
Loss factor	[HV]	0.214
Gear power loss (kW)	[PVZ]	0.211
Meshing efficiency (%)	[ηz]	97.569

14 Service life, damage

Required safety for tooth root	[S_{Fmin}]	1.4000
Required safety for tooth flank	[S_{Hmin}]	1.0000
Required service life	[H]	1000.0000

Service life (calculated with required safeties):

System service life (h)	[H_{att}]	1024

		------- Gear 1	---------- Gear 2 -------
Tooth root service life (h)	[H_{Fatt}]	1e+06	1e+06
Tooth flank service life (h)	[H_{Hatt}]	1e+06	1024

Note: The entry 1e+006 h means that the Service life > 1,000,000 h.

14.1 Damage

Damage relative to the required service life (H, 1000.0 h)

D_{F1} (%)	D_{F2} (%)	D_{H1} (%)	D_{H2} (%)
0.0000	0.0000	0.0000	97.6108

Damage relative to the system service life (H_{att}, 1024.5 h)

D_{F1} (%)	D_{F2} (%)	D_{H1} (%)	D_{H2} (%)
0.0000	0.0000	0.0000	100.0000

15 Reliability calculation

15.1 Calculation method

Calculation method according to B. Bertsche, Reliability in Automotive and Mechanical Engineering, Springer-Verlag Berlin Heidelberg 2008

15.2 Factors

Reliability of material data for σ_{Hlim} (%)	[$R\sigma_{Hlim}$]	99.00
Reliability of material data for σ_{Flim} (%)	[$R\sigma_{Flim}$]	99.00

Calculation of coefficients for reliability R(t)
$R(t) = 100 \cdot Exp(-((t \cdot fac - t_0)/(T - t_0))^{\beta})$ (%)

Gear	Type	fac	β	t_0	T	R(H)
		cycles/h		cycles	cycles	%
1	Tooth root	72000	1.700e+00	9.654e+29	1.484e+30	100.0000
1	Tooth flank	72000	1.300e+00	9.014e+29	4.295e+30	100.0000
2	Tooth root	19200	1.700e+00	9.654e+29	1.484e+30	100.0000
2	Tooth flank	19200	1.300e+00	1.773e+07	8.448e+07	99.3018

fac = Number of load cycles per hour
β = Weibull shape parameter
t_0 = Failure-free number of load cycles
T = Characteristic service life (in load cycles) for 63.2% failure probability
R(H) = Reliability for required service life

15.3 Resulting reliabilities and service lifes

Required service life (h)	[H_{min}]	1000.0000
Reliability R, tooth roots subsystem (%)	[R_{subF}]	100.0000
Reliability R, tooth flanks subsystem (%)	[R_{subH}]	99.3018
Reliability R, gears subsystem (%)	[R_{subG}]	99.3018
Required reliability (%)	[R_{min}]	99.0000
Service life H, tooth roots subsystem (h)	[H_{subF}]	> 1'000'000
Service life H, tooth flanks subsystem (h)	[H_{subH}]	1024.4771
Service life H, gears subsystem (h)	[H_{subG}]	1024.4771

16 Remarks

16.1 Conventions

- Specifications with **.e/i** mean: Maximum value **.e** and Minimum value **.i**, taking all tolerances into account.
- Specifications with **.m** mean: Mean value within tolerance.

- The center distance tolerances and the tooth thickness allowance are taken into account for the backlash tolerance. The maximum and minimum clearance corresponding to the largest and smallest allowances are shown. The calculation is performed for the operating pitch circle.

16.2 Calculations and factors

- Details of calculation method:
  c_γ according to Method B
  K_v according to Method B
  $K_{H\beta}$ and $K_{F\beta}$ according to Method C
  $f_{m\alpha}$ according to equation 66, f_{sh} according to 59/60, $F_{\beta x}$ according to 54/55/57
  $K_{H\alpha}$, $K_{F\alpha}$ according to Method B
- The logarithmically interpolated value must be entered for factors Z_L, Z_V, Z_R, Z_W, Z_X, $Y_{\delta relT}$, Y_{RrelT} and Y_X. This value is calculated from the infinite life strength and static strength values, based on the number of load cycles.

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